

2023 HKDSE Physics

Paper 2 — Elective Sections

Original questions and official answers with detailed English reasoning. Calculations show the governing relationships, substitution and units; each structured subpart is presented separately.

Editable questions with original diagrams, official answers and detailed English reasoning. Each subpart is separated for clarity.

Section A — Astronomy and Space Science

Official answers are placed directly below each question.

Question 1.1

Arrange the celestial bodies below in descending order of diameter:

- (1) the Milky Way
- (2) our Local Group of galaxies
- (3) a globular cluster of stars

A. (1), (3), (2)

B. (2), (1), (3)

C. (2), (3), (1)

D. (1), (2), (3)

Official answer: B

The Local Group contains the Milky Way and other galaxies, while a globular cluster is a much smaller collection of stars within a galaxy. Descending diameter is (2), (1), (3). Answer: B.

Question 1.2

Which of the following astronomical phenomena CANNOT be explained by Ptolemy's geocentric model?

- (1) Mercury can only be seen shortly before sunrise or shortly after sunset.
- (2) A solar eclipse can be seen on Earth when sunlight is blocked by the Moon.
- (3) Annual stellar parallax of distant stars can be seen on Earth.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

Ptolemy's model keeps the Earth stationary, so it cannot account for annual stellar parallax caused by the changing viewing position as Earth orbits the Sun. Its arrangements can account for Mercury remaining near the Sun and for the Moon blocking sunlight in a solar eclipse. Only (3) cannot be explained. Answer: B.

Question 1.3

The Tiangong space station orbits the Earth at an altitude of about 400 km above the Earth's surface. What is the gravitational acceleration of a scientist inside the space station? Given that the radius of the Earth is 6400 km and the gravitational acceleration on the Earth's surface is g .



- A. zero
- B. between zero and g , and closer to zero
- C. between zero and g , and closer to g
- D. g

Official answer: C

At orbital radius $R+h$, $g'/g = [R/(R+h)]^2 = (6400/6800)^2 = 0.886$. Gravity is therefore slightly below surface gravity, not zero. The scientist feels weightless because both scientist and station are in free fall. Answer: C.

Question 1.4

The escape velocity from planet P of radius r is v . What is the escape velocity from another planet Q of radius $2r$ and having the same mass as P?

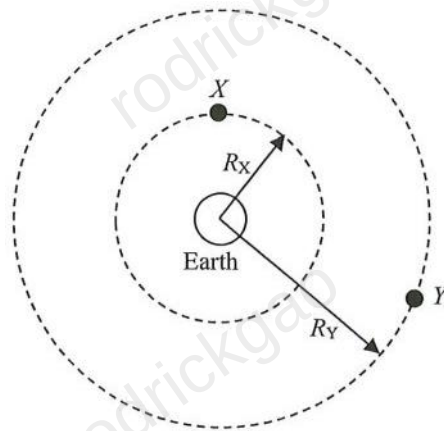
- A. $v/\sqrt{2}$
- B. v
- C. $\sqrt{2}v$
- D. It depends on the mass of the object escaping from the planet.

Official answer: A

Escape speed follows $\frac{1}{2}mv^2 = GMm/r$, so $v = \sqrt{(2GM/r)}$. Doubling the radius at unchanged planetary mass multiplies the escape speed by $1/\sqrt{2}$; the escaping object's mass cancels. Answer: A.

Question 1.5

Two satellites X and Y of the same mass orbit the Earth in circular orbits of radii R_X and R_Y respectively, where $R_X < R_Y$. Which comparison of their energies is correct?



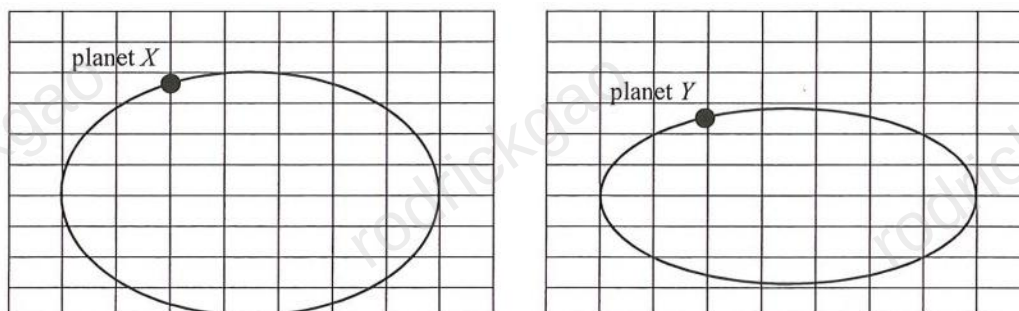
- A. Total energy: $X > Y$; kinetic energy: $X < Y$
- B. Total energy: $X > Y$; kinetic energy: $X > Y$
- C. Total energy: $X < Y$; kinetic energy: $X < Y$
- D. Total energy: $X < Y$; kinetic energy: $X > Y$

Official answer: D

For a circular orbit, $K = GMm/(2R)$ and total energy $E = -GMm/(2R)$. Since $R_X < R_Y$, X has greater kinetic energy but more negative, hence smaller, total energy. Answer: D.

Question 1.6

Two planets X and Y are equal in mass. Their orbits around stars S_x and S_y are shown. S_x and S_y are massive stars of the same mass and are not shown. Which statements are correct? (1) Their periods are different. (2) Y has a shorter semi-minor axis. (3) The shortest distance between X and S_x is different from that between Y and S_y .



- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

The diagrams show equal semi-major axes. With equal stellar masses, Kepler's third law gives equal periods, so (1) is false. Y's more elongated ellipse has a smaller semi-minor axis and a smaller periapsis distance $a(1-e)$, so (2) and (3) are true. Answer: C.

Question 1.7

The apparent and absolute magnitudes of stars X, Y and Z are tabulated below. Arrange the stars in descending order of parallax as observed from Earth.

Star	Apparent magnitude	Absolute magnitude
X	-0.8	-3.4
Y	1.3	2.7
Z	-2.7	-4.3

- A. Y, Z, X
- B. Y, X, Z
- C. X, Z, Y
- D. Z, X, Y

Official answer: A

Using $m-M = 5 \log_{10}(d/10 \text{ pc})$, the distance moduli of X, Y and Z are 2.6, -1.4 and 1.6. Distance therefore increases in the order Y, Z, X. Since parallax is inversely proportional to distance, descending parallax is Y, Z, X. Answer: A.

Question 1.8

On the Hertzsprung–Russell (H–R) diagram, the Sun is a main-sequence star. Which statements are correct?

- (1) All supergiant stars have a lower surface temperature than the Sun.
- (2) All main-sequence stars with a higher surface temperature than the Sun have a larger size than the Sun.
- (3) All white dwarfs are less luminous than the Sun.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

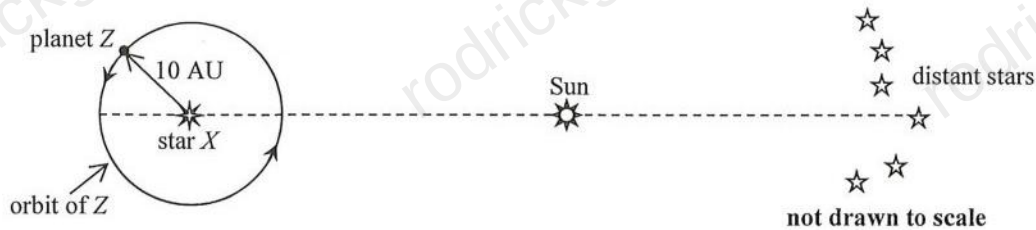
Official answer: D

Supergiants include hot blue stars as well as cool red stars, so (1) is false. On the standard H–R diagram used here, hotter main-sequence stars have larger radii than the Sun, and white dwarfs have lower luminosities despite often having high surface temperatures. Thus (2) and (3) are correct. Answer: D.

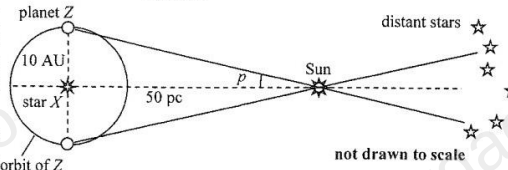
Question 1 — Structured question

Star X of surface temperature $T_x = 20\,000\text{ K}$ is a main-sequence star that shines with a luminosity 1000 times that of the Sun.

- (a) Given that the surface temperature of the Sun is 5800 K , find the radius of X in terms of the solar radius R_s . (2 marks)
- (b) Star Y in the main sequence has the same apparent magnitude as X. A student thought that with this information one can determine whether the surface temperature of Y is equal to, greater than or smaller than T_x . Do you agree? Explain. (3 marks)
- (c) The wavelength of a hydrogen emission line measured in the laboratory is 486.1 nm . In the observed spectrum of X, the same spectral line is detected at 485.7 nm . Determine the motion of X relative to the Earth and estimate its minimum speed in km s^{-1} . (3 marks)
- (d) The Sun is at a distance of 50 pc from X. A planet Z orbits X in a circular orbit of radius 10 AU . Figure 1.1 (not drawn to scale) shows X, Z, the Sun and some distant stars. Suppose that an observer on Z uses the parallax method to measure the distance between X and the Sun. Calculate the parallax angle p of the Sun in arcseconds. (2 marks)



Official marking scheme

Solution	Marks	Remarks
1. (a) By Stefan's law, $L_S = \sigma(4\pi R_S^2)T_S^4$ $L_X = \sigma(4\pi R_X^2)T_X^4$ where L_S, R_S, and T_S represent the luminosity, the radius and the surface temperature of the Sun respectively, while L_X, R_X, and T_X represent the respective quantities of star X. $\frac{L_X}{L_S} = \left(\frac{R_X}{R_S}\right)^2 \left(\frac{T_X}{T_S}\right)^4$ On solving, $R_X = \sqrt{(1000) \left(\frac{5800}{20000}\right)^4} R_S$ $= 2.659476 R_S \approx 2.66 R_S$	1M 1A 2	Accept (2.65 ~ 2.67) R_S
(b) It is indeterminate. Although the two stars have the same apparent magnitude, their distances from Earth are unknown and so one cannot determine which star has a greater absolute magnitude (i.e. their absolute magnitudes may be equal or different). As absolute magnitude determines the luminosity / surface temperature, the surface temperature of star Y may be higher, lower or equal to that of star X.	1A 1A 1A 3	
(c) Star X is approaching the Earth with a minimum speed of $v = \frac{\lambda_{\text{obs}} - \lambda_0}{\lambda_0} \times c = \frac{-0.4}{486.1} \times 3.0 \times 10^8$ $= -246.863 \text{ (km s}^{-1}\text{)} \approx -246.9 \text{ (km s}^{-1}\text{)}$ Negative sign means star X is approaching the Earth.	1M 1A 1A 3	Accept 246 ~ 247 (km s⁻¹)
(d) The parallax angle p [arcsec] = $\frac{r[\text{AU}]}{D[\text{parsec}]}$, where r is the orbital radius in units of AU and D is the distance to the Sun in units of parsec. Thus, $p = \frac{10[\text{AU}]}{50[\text{parsec}]} = 0.2$ (arcseconds). 	1M 1A 2	Note: $\frac{0.2\pi}{3600 \times 180^\circ} = 9.696 \times 10^{-7} \text{ rad}$

(a) Stefan's law gives $L = 4\pi R^2 \sigma T^4$. Thus $R_X/R_S = \sqrt{(L_X/L_S)(T_S/T_X)^2} = \sqrt{1000(5800/20000)^2} = 2.66$. The radius is $2.66 R_S$.

(b) No. Equal apparent magnitudes mean equal received brightness, not equal luminosity. The distances are unknown, so the absolute magnitudes and luminosities cannot be compared. Even though both stars are on the main sequence, Y may have a higher, lower or equal surface temperature.

(c) $\Delta\lambda = 485.7 - 486.1 = -0.4 \text{ nm}$, a blueshift, so X approaches Earth. Its radial velocity is $v_r = c\Delta\lambda/\lambda_0 = (3.0 \times 10^5)(-0.4)/486.1 = -246.9 \text{ km s}^{-1}$. The minimum speed is 246.9 km s^{-1} ; any transverse component increases the actual speed.

(d) The parallax baseline radius is 10 AU, not the full 20 AU diameter. Therefore $p = (10 \text{ AU})/(50 \text{ pc}) = 0.20$ arcsec.

Section B — Atomic World

Question 2.1

In a Rutherford scattering experiment, most α -particles pass straight through gold foil while a few are deflected significantly. This shows that an atom has a tiny massive nucleus. Suppose the mass of an atom is evenly distributed within the whole atom.

- A. most α -particles would be stopped by the gold foil.
- B. nearly all α -particles would pass straight through the gold foil.
- C. most α -particles would be deflected significantly and only a few would pass straight through the gold foil.
- D. most α -particles would rebound by 180° .

Official answer: B

Large-angle Rutherford scattering requires a highly concentrated massive scattering centre. If the atomic mass were spread through the whole atom, there would be no tiny massive nucleus producing these large deflections, and nearly all alpha particles would continue approximately straight through. Answer: B.

Question 2.2

Which statements about atomic spectra are correct?

- (1) Atomic spectra are due to discrete energy levels for electrons in atoms.
- (2) Absorption lines in atomic spectra arise when orbital electrons gain energy.
- (3) Emission lines in atomic spectra only correspond to the visible part of the electromagnetic spectrum.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

Discrete electron energy levels give photons of specific energies, $\Delta E = hf$. Absorption raises an electron to a higher energy level, so (1) and (2) are true. Downward transitions can emit ultraviolet or infrared as well as visible photons, making (3) false. Answer: A.

Question 2.3

A hydrogen atom at ground state has an ionization energy of 13.6 eV. Which of the following photons can be absorbed by this hydrogen atom?

- (1) 3.4 eV
- (2) 10.2 eV
- (3) 13.7 eV

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

The first excitation requires $13.6(1-1/2^2) = 10.2$ eV, so a 10.2 eV photon can be absorbed but a 3.4 eV photon cannot excite the ground-state atom. A 13.7 eV photon exceeds the 13.6 eV ionization energy and can ionize it, leaving about 0.1 eV as electron kinetic energy. Thus (2) and (3) apply. Answer: C.

Question 2.4

Monochromatic light illuminates the cathodes of photocells X and Y, each connected to a sensitive galvanometer. The galvanometer connected to X shows a reading while that connected to Y does not. Which statements are correct?

- (1) Electrons can escape from the cathode of X but not from that of Y.
- (2) Compared with Y, the metallic cathode of X has a smaller threshold frequency.
- (3) Compared with Y, the metallic cathode of X has a larger work function.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

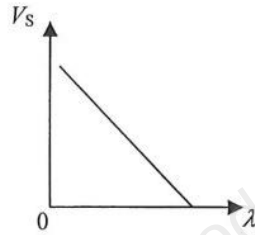
Official answer: A

At the same incident frequency, photoemission occurs from X but not Y, so the photon energy exceeds X's work function but is insufficient for Y. Hence $f_{0,X} < f_{0,Y}$ and $\phi_X < \phi_Y$. Statements (1) and (2), not (3), are correct. Answer: A.

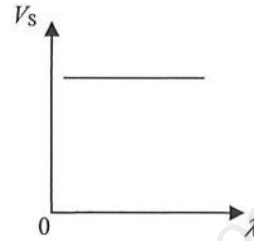
Question 2.5

In a photoelectric experiment, monochromatic lights of different wavelength λ are incident on the cathode of the same photocell and the corresponding stopping potential V_s is measured. Which graph of V_s against λ is correct?

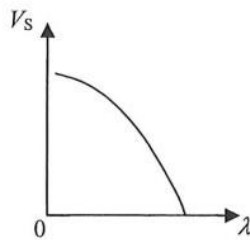
A.



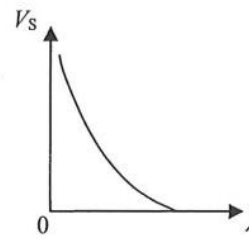
B.



C.



D.



A. Graph A

B. Graph B

C. Graph C

D. Graph D**Official answer: D**

Einstein's equation gives $eV_s = hc/\lambda - \phi$, so $V_s = hc/(e\lambda) - \phi/e$. As λ increases, the stopping potential decreases along a curve that becomes less steep, reaching zero at the threshold wavelength. This is graph D; beyond threshold there is no photoemission. Answer: D.

Question 2.6

An electron has a de Broglie wavelength of 0.10 nm. Estimate its average kinetic energy.

A. 151 eV

B. 302 eV

C. 6.2 keV

D. 12.4 keV

Official answer: A

With $\lambda = 0.10 \text{ nm} = 1.0 \times 10^{-10} \text{ m}$, $p = h/\lambda$ and $K = p^2/(2m_e) = h^2/(2m_e\lambda^2)$. Substitution gives $K = (6.63 \times 10^{-34})^2/[2(9.11 \times 10^{-31})(1.0 \times 10^{-10})^2] = 2.41 \times 10^{-17} \text{ J} = 151 \text{ eV}$. Answer: A.

Question 2.7

The angular resolution limit of a transmission electron microscope is 4×10^{-10} rad when the accelerating voltage is 5 kV. Estimate the angular resolution limit when the voltage is 20 kV. Assume that the resolution is limited by diffraction only.

- A. 1×10^{-10} rad
- B. 2×10^{-10} rad
- C. 8×10^{-10} rad
- D. 16×10^{-10} rad

Official answer: B

For non-relativistic electrons, $eV = p^2/(2m_e)$, so $\lambda = h/p \propto 1/\sqrt{V}$. With the same aperture and diffraction-limited resolution, $\theta \propto \lambda$. Raising voltage from 5 to 20 kV therefore halves the angular limit to 2×10^{-10} rad. Answer: B.

Question 2.8

Which statements illustrate nanoscale surface effects?

- (1) Nanosized titanium dioxide is an efficient catalyst.
- (2) Nanosized silver particles have a lower melting point than bulk silver.
- (3) Current can flow across a nanoscale gap even when there is no physical contact.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

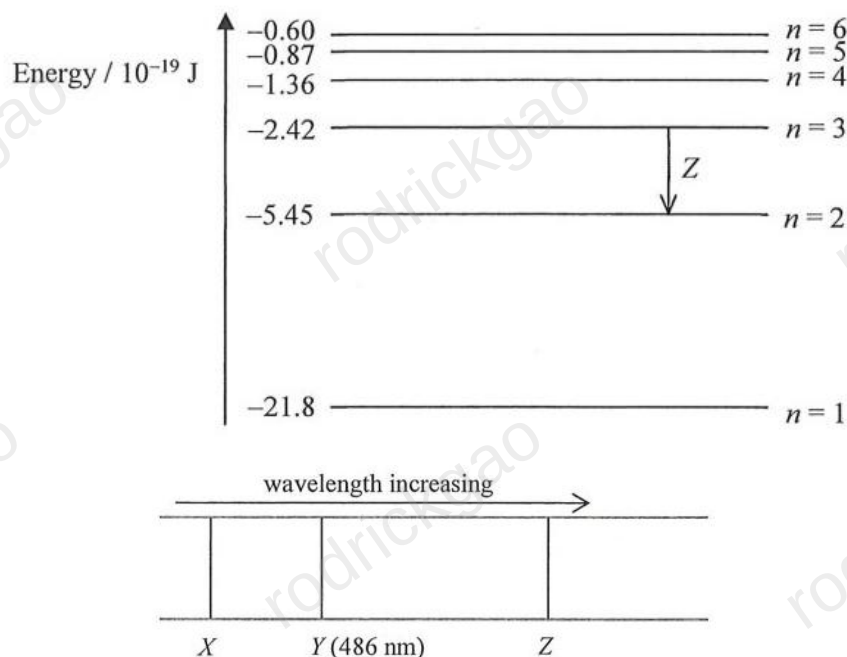
Nanoparticles have a large surface-area-to-volume ratio and a large fraction of atoms at the surface. This enhances catalytic action and can reduce melting point, explaining (1) and (2). Current across a gap is quantum tunnelling, a quantum-size effect rather than the surface effect tested here. Answer: C.

Question 2 — Structured question

The Bohr model explains the discrete spectrum of hydrogen.

- (a) One postulate is $m_e v r = n h / (2\pi)$, $n = 1, 2, 3, \dots$, where h is the Planck constant, m_e is the electron mass, v is its speed and r is the orbital radius. Given that the radius of the n th orbit is $r_n = n^2 a_0$, where $a_0 = 5.29 \times 10^{-11}$ m.
- (i) Show that the electron's de Broglie wavelength is quantized and can be written as $\lambda = n \times \text{constant}$. (1 mark)
- (ii) Find the wavelength of the electron in the innermost orbit. (1 mark)

Figure 2.1 shows the energy levels of a hydrogen atom. A ground-state hydrogen atom is illuminated by monochromatic radiation. Only three visible spectral lines X, Y and Z are observed; line Y has wavelength 486 nm. The energy of a visible photon lies between 2.84×10^{-19} J and 4.97×10^{-19} J.



- (b) (i) Transition Z is from $n = 3$ to $n = 2$. State the transition that produces X. (1 mark)
- (ii) Calculate the wavelength of X. (2 marks)
- (iii) Determine whether a single transition from $n = 6$ gives visible light. Hence state and explain the energy level to which the atom was excited. (3 marks)
- (c) Find the energy of the monochromatic radiation in eV and state whether it is infrared, visible or ultraviolet. (2 marks)

Official marking scheme

Solution	Marks	Remarks
2. (a) (i) $m_e v r = n \frac{h}{2\pi}$ $2\pi r = n \frac{h}{m_e v} = n\lambda$ $2\pi r_n = 2\pi n^2 a_0 = n\lambda_n$ $\lambda_n = n \times 2\pi a_0$	1A 1	
(ii) $\lambda_1 = 1 \times 2\pi a_0 = 2\pi \times 5.29 \times 10^{-11}$ $= 3.323805 \times 10^{-10} \text{ m} \approx 3.32 \times 10^{-10} \text{ m}$	1A 1	Accept $(3.30 \sim 3.33) \times 10^{-10} \text{ m}$
(b) (i) X: $n = 5$ to $n = 2$ (i.e. 4 th excited state to 1 st excited state)	1A 1	
(ii) $\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{(-0.87 - (-5.45)) \times 10^{-19}} = 4.34279476 \times 10^{-7} \text{ m}$ $\approx 434 \text{ nm}$	1M 1A 2	Accept $(432 \sim 434) \text{ nm}$
(iii) Transition from $n = 6$ to $n = 2$ gives $\Delta E = (-0.60 - (-5.45)) \times 10^{-19} \text{ J}$ $= 4.85 \times 10^{-19} \text{ J}$ (within visible) There is a transition that gives a visible spectral line.	1M 1A	
Or $\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{(-0.60 - (-5.45)) \times 10^{-19}} = 410.103093 \text{ nm} \approx 410 \text{ nm}$ (within visible)	1M 1A	
Or For a transition from 6 to n , if the emitted photon is visible, $4.97 \times 10^{-19} \geq (-0.60 \times 10^{-19} - E_n) \geq 2.84 \times 10^{-19}$ $-3.44 \times 10^{-19} \geq E_n \geq -5.57 \times 10^{-19}$ $n = 2$ ($E_n = -5.45 \times 10^{-19} \text{ J}$) fulfill this requirement.	1M 1A	
However, no such visible emission line is found beyond the shortest visible line X. Thus, the excitation by the monochromatic radiation is from $n = 1$ to $n = 5$.	1A 3	
(c) photon energy $= [-0.87 - (-21.8)] \times 10^{-19} \text{ J}$ $= 20.93 \times 10^{-19} \text{ J} = 13.08125 \text{ (eV)} \approx 13.1 \text{ (eV)}$ ultra-violet (UV)	1A 1M/1A 2	Accept $(13.0 \sim 13.1) \text{ (eV)}$

(a)(i) From $m_e v r_n = nh/(2\pi)$ and $\lambda_n = h/(m_e v)$, $2\pi r_n = n\lambda_n$. With $r_n = n^2 a_0$, $\lambda_n = 2\pi n a_0 = n \times \text{constant}$, demonstrating quantization.

(a)(ii) $\lambda_1 = 2\pi a_0 = 2\pi(5.29 \times 10^{-11}) = 3.32 \times 10^{-10} \text{ m}$.

(b)(i) X is the transition $n = 5 \rightarrow n = 2$ (fourth excited state to first excited state).

(b)(ii) Using the tabulated energy levels, $\Delta E = [-0.87 - (-5.45)] \times 10^{-19} = 4.58 \times 10^{-19} \text{ J}$. Therefore $\lambda_X = hc/\Delta E = (6.63 \times 10^{-34})(3.0 \times 10^8)/(4.58 \times 10^{-19}) = 4.34 \times 10^{-7} \text{ m} = 434 \text{ nm}$.

(b)(iii) The transition $n = 6 \rightarrow n = 2$ releases $[-0.60 - (-5.45)] \times 10^{-19} = 4.85 \times 10^{-19} \text{ J}$, within the given visible-photon range. Its wavelength is about 410 nm, shorter than X. No such fourth visible line is observed, so the initial excitation is from $n = 1$ to $n = 5$, not to $n = 6$ or above.

(c) The incident photon energy is $E_5 - E_1 = [-0.87 - (-21.8)] \times 10^{-19} = 2.093 \times 10^{-18} \text{ J} = 13.1 \text{ eV}$. This radiation is ultraviolet.

Section C — Energy and Use of Energy

Question 3.1

The energy label of a 12 W compact fluorescent lamp is shown. Which statements are correct?

- (1) Its luminous efficacy is 63 lm W^{-1} .
- (2) The illuminance of the lamp is 756 lx.
- (3) Compared with other 12 W compact fluorescent lamps, its efficacy is above average.



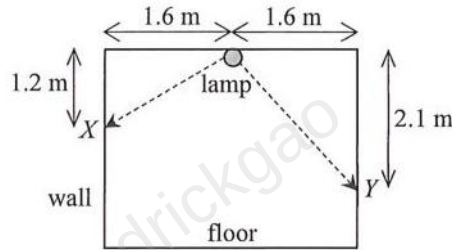
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

The label states a luminous efficacy of 63 lm W^{-1} . At 12 W, the corresponding luminous flux is $63 \times 12 = 756 \text{ lm}$, not an illuminance of 756 lx. Lumens are not lux: illuminance also depends on position and geometry, so (2) does not follow. Grade 1 indicates above-average efficiency among the comparable 12 W compact fluorescent lamps. Thus (1) and (3) are correct. Answer: B.

Question 3.2

A lamp is installed at the centre of a square ceiling, 1.6 m horizontally from each wall. X is on one wall 1.2 m below the ceiling, and Y is on the opposite wall 2.1 m below the ceiling, as shown. Neglecting reflection, estimate the ratio of illuminance at X to that at Y.



- A. 0.43
- B. 0.76
- C. 1.32
- D. 2.30

Official answer: D

For a point source illuminating a vertical wall, $E = I \cos\theta/r^2$. Here $r_X = \sqrt{(1.6^2 + 1.2^2)} = 2.00$ m and $r_Y = \sqrt{(1.6^2 + 2.1^2)} = 2.64$ m. The wall normals are horizontal, so $\cos\theta = 1.6/r$. Therefore $E_X/E_Y = (r_Y/r_X)^3 = (2.64/2.00)^3 = 2.30$. The inverse-square law alone would omit the incidence-angle factor. Answer: D.

Question 3.3

The heat-transfer data for the external walls, roof and windows of a building are tabulated. The overall thermal transfer value is 22.4 W m^{-2} . Find the total window area x .

Component	Area / m^2	Heat-transfer rate / W m^{-2}
External walls	102	10
Roof	70	25
Windows	x	116

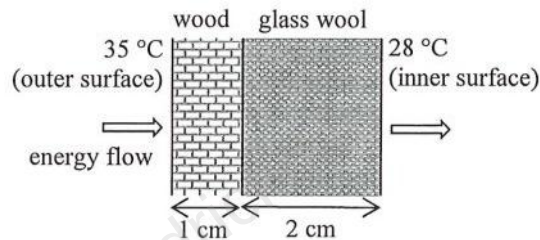
- A. 7.8 m^2
- B. 9.3 m^2
- C. 11.6 m^2
- D. 23.9 m^2

Official answer: C

OTTV is total heat-gain rate divided by total envelope area: $[102(10) + 70(25) + 116x] / (102 + 70 + x) = 22.4$. Rearranging, $93.6x = 1082.8$, hence $x = 11.6 \text{ m}^2$. Answer: C.

Question 3.4

A wall consists of 1 cm of wood and 2 cm of glass wool. The outer and inner surface temperatures are 35 °C and 28 °C. The thermal conductivities of wood and glass wool are 0.15 and 0.03 W m⁻¹ K⁻¹ respectively. Find the temperature at their interface.



- A. 28.6 °C
- B. 30.0 °C
- C. 33.0 °C
- D. 34.4 °C

Official answer: D

In steady state the same heat flux crosses both layers. If T is the interface temperature, $(0.15/0.01)(35-T) = (0.03/0.02)(T-28)$. Thus $15(35-T) = 1.5(T-28)$, giving $T = 34.4$ °C. Most of the temperature drop occurs across the more insulating glass wool. Answer: D.

Question 3.5

An electric vehicle of mass 1500 kg slows from 70 km h⁻¹ to 35 km h⁻¹ by regenerative braking. If 57.6 kJ is stored, estimate the efficiency.

- A. 22%
- B. 25%
- C. 27%
- D. 30%

Official answer: C

The kinetic-energy decrease is $\Delta K = \frac{1}{2}(1500)[(70/3.6)^2 - (35/3.6)^2] = 2.13 \times 10^5$ J. Efficiency = stored chemical energy/kinetic-energy decrease = $57.6 \times 10^3 / (2.13 \times 10^5) = 0.271 \approx 27\%$. Use the decrease in kinetic energy, not the initial energy alone. Answer: C.

Question 3.6

Solar panels usually have a dark appearance. This is because

- A. the cost of manufacturing dark-coloured solar panels is lower.
- B. a transparent anti-reflection film is coated onto the top surface of the panel.
- C. the panels are painted dark so as to absorb more radiation.
- D. the panels are painted dark so that they need not be cleaned as frequently.

Official answer: B

A transparent anti-reflection coating reduces the light reflected from the panel, allowing more light to enter the photovoltaic material. With less visible light reflected to an observer, the panel appears dark. It is not painted opaque black. Answer: B.

Question 3.7

There are 100 wind turbines, each with blade length 6 m and efficiency 23%. They are replaced by N newer turbines with blade length 8 m and efficiency 28%. Estimate N, assuming the total output power is almost unchanged.

- A. 46
- B. 62
- C. 69
- D. 91

Official answer: A

At the same wind speed, useful turbine power is proportional to efficiency times swept area: $P \propto \eta \pi r^2$. Equating the two total outputs gives $N(0.28)(8^2) = 100(0.23)(6^2)$, so $N = 46.2 \approx 46$ turbines. Answer: A.

Question 3.8

Which statements about nuclear energy are correct?

- (1) It is a kind of renewable energy.
- (2) If the magnitude of total binding energy increases after a nuclear reaction, energy is released.
- (3) Unstable nuclei tend to release energy through nuclear reactions to become more stable.

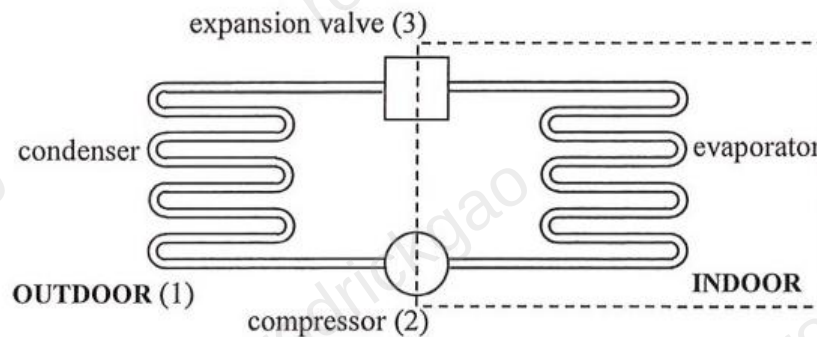
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

Conventional nuclear fuel is finite and is not classified as renewable, so (1) is false. A greater total binding-energy magnitude means lower total rest energy and energy released, so (2) is true. Energy-releasing nuclear changes can take unstable nuclei towards more stable products, making (3) true. Answer: D.

Question 3 — Structured question

Figure 3.1 shows the components of an air-conditioner and the direction in which refrigerant circulates.



- (a) (i) State the changes of state of the refrigerant and explain how heat is removed from indoors and released outdoors. (2 marks)
- (ii) Which has the highest temperature: outdoor air, refrigerant leaving the compressor, or refrigerant leaving the expansion valve? (1 mark)
- (iii) Explain why the heat Q_H released outdoors is greater than the heat Q_C removed from indoors. (1 mark)
- (b) The air-conditioner removes $2.04 \times 10^6 \text{ J}$ from a classroom in 4.0 min.
- (i) Find its cooling capacity C_p . (2 marks)
- (ii) The classroom measures $13.4 \text{ m} \times 5.0 \text{ m} \times 3.0 \text{ m}$. Estimate the decrease in air temperature. The specific heat capacity and density of air are $1000 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$ and 1.20 kg m^{-3} . (2 marks)
- (iii) If the coefficient of performance is 6.2, estimate the total heat Q_H released. (2 marks)

Official marking scheme

Solution		Marks	Remarks
(a) (i)	At the evaporator indoor, liquid refrigerant evaporates and absorbs latent heat from the air indoor. At the condenser outdoor, vapour refrigerant condenses and releases latent heat to the air outdoor.	1A	
		1A	
		2	
(ii)	(2): leaving the compressor	1A	
		1	
(iii)	Energy consumed by the air-conditioner / work W has to be done on the refrigerant, and it is released in addition to the heat removed Q_C . (i.e. by conservation of energy, $Q_H = Q_C + W$.)	1A	
(b) (i)	$Q_C = 2.04 \times 10^6 = C_p \times 4 \times 60$ $C_p = 8500 \text{ W} = 8.50 \text{ kW}$	1M	
		1A	
(ii)	$Q_C = mc\Delta\theta$ $2.04 \times 10^6 = (1.20 \times 13.4 \times 5.0 \times 3.0) \times 1000 \times \Delta\theta$ $\Delta\theta = 8.457711^\circ\text{C} \approx 8.46^\circ\text{C}$	1M	Accept $(8.4 \sim 8.5)^\circ\text{C}$
		1A	
(iii)	$W = Q_C \div 6.2$ $Q_H = Q_C + W = Q_C + \frac{Q_C}{6.2} = (2.04 \times 10^6) \times (1 + \frac{1}{6.2})$ $= 2.369032 \times 10^6 \text{ J} \approx 2.37 \times 10^6 \text{ J}$	1M	Accept $(2.35 \times 10^6 \sim 2.40 \times 10^6) \text{ J}$
		1A	
Or Electrical power input $P = \frac{C_p}{6.2}$ $Q_H / t = C_p + P = C_p + \frac{C_p}{6.2} = 8500 \times (1 + \frac{1}{6.2})$ $= 9870.9677 \text{ W} \approx 9871 \text{ W}$ $Q_H = 9871 \times (4 \times 60) = 2.369032 \times 10^6 \text{ J} \approx 2.37 \times 10^6 \text{ J}$		1M	
		1A	
			2

(a)(i) In the indoor evaporator, liquid refrigerant evaporates and absorbs latent heat from the indoor air. In the outdoor condenser, refrigerant vapour condenses and releases latent heat to the outdoor air.

(a)(ii) Temperature (2), that of the refrigerant leaving the compressor, is the highest.

(a)(iii) Work W is supplied to drive the compressor. By energy conservation, $Q_H = Q_C + W$, so the heat released outdoors exceeds that removed indoors.

(b)(i) Cooling capacity $C_p = Q_C/t = (2.04 \times 10^6)/(4.0 \times 60) = 8500 \text{ W} = 8.50 \text{ kW}$.

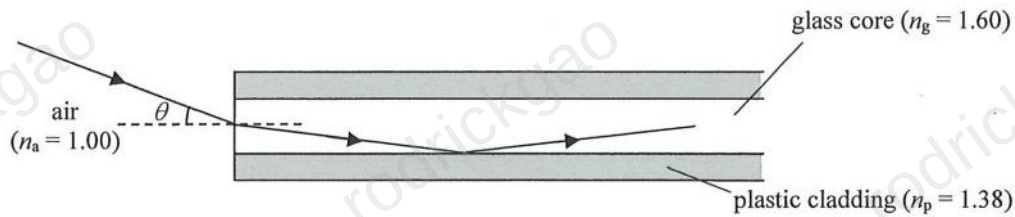
(b)(ii) Air mass $m = \rho V = 1.20(13.4 \times 5.0 \times 3.0) = 241.2 \text{ kg}$. From $Q_C = mc\Delta T$, $\Delta T = (2.04 \times 10^6)/(241.2 \times 1000) = 8.46^\circ\text{C}$. This estimate neglects heat entering the room and heat stored in its contents.

(b)(iii) $\text{COP} = Q_C/W = 6.2$, hence $Q_H = Q_C + Q_C/6.2 = (2.04 \times 10^6)(1 + 1/6.2) = 2.37 \times 10^6 \text{ J}$.

Section D — Medical Physics

Question 4.1

An optical fibre has a glass core of refractive index 1.60 and plastic cladding of refractive index 1.38; the refractive index of air is 1.00. For total internal reflection in the core, what is the allowed range of incident angle θ ?



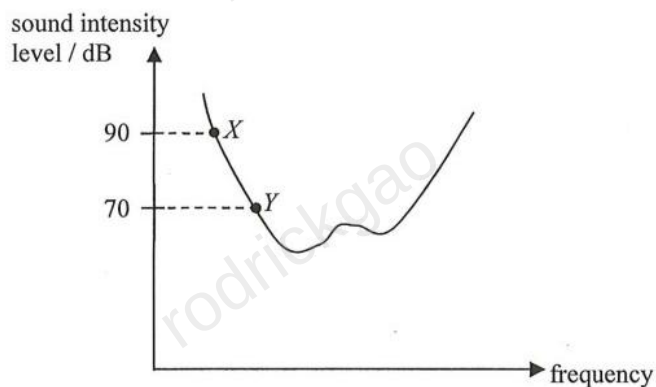
- A. θ smaller than 54.1°
- B. θ larger than 54.1°
- C. θ smaller than 59.6°
- D. θ larger than 59.6°

Official answer: A

At the core–cladding interface, $c = \sin^{-1}(1.38/1.60) = 59.6^\circ$. If the refracted ray makes angle r with the fibre axis, its incidence at that interface is $90^\circ - r$; total internal reflection needs $r < 30.4^\circ$. At entry, $\sin\theta = 1.60 \sin r$, giving $\theta < \sin^{-1}[\sqrt{(1.60^2 - 1.38^2)}] = 54.1^\circ$. Do not confuse the entry angle with the internal critical angle. Answer: A.

Question 4.2

On an equal-loudness contour, sounds X and Y both lie on the 70-phon curve. Their sound intensity levels are 90 dB and 70 dB respectively. Which statement is correct?



- A. X is 20 dB louder than Y.
- B. Y is louder than X.
- C. They are equally loud, but the intensity of X is 20 times that of Y.
- D. They are equally loud, but the intensity of X is 100 times that of Y.

Official answer: D

Both notes lie on the same 70-phon contour, so they are equally loud. Their intensity-level difference is $90 - 70 = 20$ dB. Since $\Delta L = 10 \log_{10}(I_X/I_Y)$, the intensity ratio is $10^2 = 100$, not 20. Answer: D.

Question 4.3

The table shows typical frequencies and properties of ultrasound used in scans. Which statements are correct?

- (1) Higher-frequency ultrasound gives better resolution.
- (2) Large penetration depth and high resolution can be achieved at a certain frequency.
- (3) To investigate the whole liver, a 5–10 MHz scan is best.

Frequency / MHz	Typical penetration depth	Resolution along beam / mm
3–5	10–20 cm	1.0
5–10	5 cm	0.2
10–15	1 cm	0.1
50	a few mm	0.05

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: A

The table shows that higher frequency gives a smaller resolvable separation and hence better resolution, so (1) is true. It also reduces penetration depth, contradicting (2). The 5–10 MHz beam penetrates only about 5 cm, insufficient for the whole liver; the lower 3–5 MHz band offers the greater depth needed. Only (1) is correct.
Answer: A.

Question 4.4

Which statement about computed tomography (CT) is INCORRECT?

- A. Grey levels in CT images correspond to the X-ray attenuation coefficients of body tissues.
- B. CT image reconstruction involves back-projecting intensity readings of X-ray beams across an image plane viewed at different angles.
- C. CT scans provide both anatomical and functional information of an organ.
- D. The radiation dose received by a patient in a CT scan is higher than that in conventional X-ray imaging.

Official answer: C

In the conventional CT comparison used in this question, CT primarily provides anatomical information from differences in X-ray attenuation; functional imaging is associated with techniques such as radionuclide imaging. Thus statement C is the incorrect one. The other statements describe CT attenuation contrast, angular reconstruction and its generally higher dose than a conventional radiograph. Answer: C.

Question 4.5

Which imaging methods are NOT SUITABLE for guiding a needle accurately as it is inserted into the body?

- (1) ultrasound scan
- (2) endoscopy
- (3) radionuclide imaging

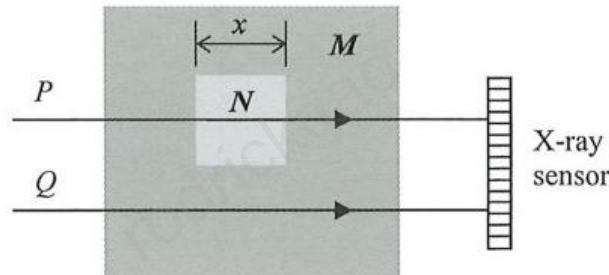
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: D

Ultrasound provides real-time cross-sectional images suitable for following a needle in tissue. Conventional endoscopy views accessible internal surfaces rather than the surrounding tissue cross-section, while radionuclide imaging lacks the spatial detail needed for accurate needle guidance. Thus the unsuitable methods in this comparison are (2) and (3). Answer: D.

Question 4.6

Two identical X-ray beams P and Q pass through the same thickness of tissue. A mass N of thickness x is embedded in tissue M in the path of P. The linear attenuation coefficients of M and N are μ_1 and μ_2 . The sensor detects intensity I where Q falls. Deduce the corresponding intensity of P.



- A. $I \exp(-\mu_2 x)$
- B. $I \exp[-(\mu_2 - \mu_1)x]$
- C. $I \exp[-(\mu_2 + \mu_1)x]$
- D. $I \exp(\mu_1 x) - I \exp(\mu_2 x)$

Official answer: B

Let the common path thickness be L . Beam Q gives $I = I_0 \exp(-\mu_1 L)$. For P, thickness x of M is replaced by N, so $I_P = I_0 \exp[-\mu_1(L-x) - \mu_2 x] = I \exp[-(\mu_2 - \mu_1)x]$. The relevant attenuation is the difference between coefficients. Answer: B.

Question 4.7

With the aid of a contrast medium, the X-ray radiographic image shown is obtained. Which statement is INCORRECT?



- A. The contrast medium absorbs X-rays and looks white on the image.
- B. The contrast medium enhances tissue and structure absorption differentials.
- C. Elements of low atomic number are used for the contrast medium.
- D. Low-energy X-rays are used to produce the image.

Official answer: C

An X-ray contrast medium normally contains high-atomic-number elements so that it attenuates X-rays more strongly than surrounding soft tissue. It then appears white and improves absorption contrast. The claim that low-atomic-number elements are used is therefore incorrect. Answer: C.

Question 4.8

A tracer labelled with ^{125}I is used in radionuclide imaging. Its physical half-life is 60 days and its effective half-life is 16 days. Estimate its biological half-life in the body.

- A. 12 days
- B. 22 days
- C. 38 days
- D. 44 days

Official answer: B

Physical decay and biological removal both reduce tracer activity: $1/T_e = 1/T_p + 1/T_b$. Therefore $1/T_b = 1/16 - 1/60$ and $T_b = 21.8 \text{ days} \approx 22 \text{ days}$. The effective half-life must be shorter than either individual half-life. Answer: B.

Question 4 — Structured question

(a) Describe the differences between rods and cones in their functions in vision. (2 marks)

The table gives the change of near point with age. At all ages the person can view distant objects clearly, and the separation between the eye lens and the retina remains 2.5 cm.

Age / year	Near point / cm
20	10
30	14
40	25
50	40
60	80
70	100

(b) (i) State the meaning of near point. (1 mark)

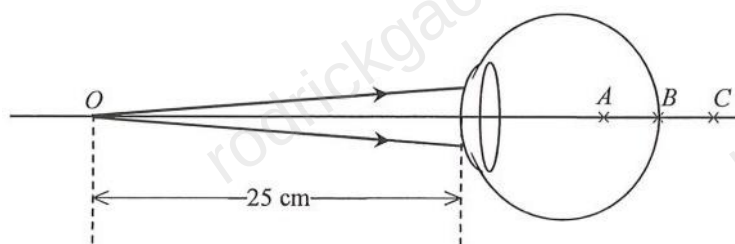
(ii) Explain why the power of the eye decreases with age. (1 mark)

(iii) State what is meant by accommodation. (1 mark)

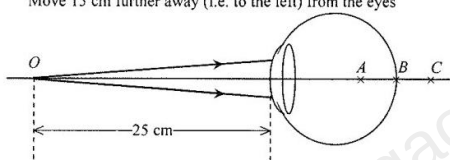
(iv) (I) Find the range of powers of the eye, in dioptres, between the far point and near point at age 20.

(II) By how much does this range decrease when the person reaches age 70? (3 marks)

(c) A 50-year-old person looks at O on a newspaper 25 cm from the eye, as shown in Figure 4.1. State whether the rays should converge at A, B or C, and state the direction in which the newspaper should be moved and the least distance required for clear vision. (2 marks)



Official marking scheme

Solution	Marks	Remarks
4. (a) Rods do not register colour but are more sensitive to light than cones so can operate at lower light intensities (thus for scotopic (dim light or night time) vision). Cones differentiate colour as they contain three visual pigments. They need higher light intensities than rods (thus for photopic (bright or daylight) vision).	1A 1A 2	
(b) (i) It is the closest point / shortest (object) distance at which an eye can focus (comfortably) / see a clear / sharp image.	1A 1	
(ii) The lens loses its flexibility and reduces its refractive power (accommodation) for focusing an object at near distance / Or The weakening (muscles around eye's lens) ciliary muscles that control / contract the lens thickness.	1A 1A 1	
(iii) Accommodation of the eyes refers to the ability of the eyes to focus / change focal length / change thickness of lens at different distances (from near point to far point).	1A 1	
(iv) The range of accommodation Δ_{20} at 20 years old: At far point, P_f power of eyes = $\frac{1}{0.025} = +40$ D At near point, P_n power of eyes = $\left(\frac{1}{0.10} + \frac{1}{0.025}\right) = +50$ D Δ_{20} is from +40 D to +50 D, i.e. $\Delta_{20} = 10$ D The range of accommodation changes, Δ_{70} at 70 years is decreased by considering the change of near point (assume the far point is fixed at 20 and 70 years old) $+50$ D - $\left(\frac{1}{1.00} + \frac{1}{0.025}\right) = +50$ D - (+41 D) = 9 D Or $\Delta = \Delta_{20} - \Delta_{70} = \frac{1}{0.10} - \frac{1}{1.00} = 9$ D	1M 1A 1A 1A 3	Note: Δ_{70} at 70 years old is narrowed to +40 D $\rightarrow$ +41 D, i.e. $\Delta_{70} = 1$ D only
(c) Point C (behind the retina) Move 15 cm further away (i.e. to the left) from the eyes 	1A 1A 2	

(a) Rods are more sensitive to light and provide dim-light vision without colour discrimination. Cones operate best in brighter light and enable colour vision.

(b)(i) The near point is the closest point at which the eye can focus to form a clear image.

(b)(ii) With age the eye lens loses flexibility, reducing its ability to become sufficiently curved and increase its refractive power for near objects.

(b)(iii) Accommodation is the eye's ability to change lens curvature and focal length so that objects at different distances are focused on the retina.

(b)(iv)(I) With retinal distance $v = 0.025$ m, the far-point power is $P_f = 1/v = 40$ D. At age 20 the near point is 0.10 m, so $P_n = 1/0.10 + 1/0.025 = 50$ D. The power range is 40–50 D, a span of 10 D.

(b)(iv)(II) At age 70 the near point is 1.00 m, giving $P_n = 1/1.00 + 1/0.025 = 41$ D. The span is then 1 D, so the accommodation range decreases by $10 - 1 = 9$ D.

(c) At age 50 the near point is 40 cm, so a newspaper at 25 cm is too close and the rays would converge at C, behind the retina. Move the newspaper away from the eye (left in the diagram) by at least $40 - 25 = 15$ cm.